

Robust Trust with Multiple Advisers

Victor Xucheng Chen

School of Management and Economics
The Chinese University of Hong Kong, Shenzhen

for GAAC 2026

The full paper is available on my personal website.

Motivating example

Three AI advisers assess the same evidence about a firm's project.

	Adviser A	Adviser B	Adviser C
Reported success probability	80%	80%	20%

Should the firm follow the majority, or average the reports?

Motivating example

Three AI advisers assess the same evidence about a firm's project.

	Adviser A	Adviser B	Adviser C
Reported success probability	80%	80%	20%

Should the firm follow the majority, or average the reports?

Two possible explanations

Common posterior	Advisers A and B	Adviser C
80%	Truthful reports	False report
20%	Coordinated false reports	Truthful report

The two explanations need not be equally likely.

The aggregation problem

Common evidence

Additional advisers do not bring additional signals. Their potential value comes from how their reports can be used.

Unknown reporting behavior

Misaligned advisers may choose their reports jointly. Agreement can reflect either common evidence or a coordinated false report.

Evaluate the aggregation rule by its worst-case expected payoff.

Related literature

Single-adviser robust trust

Dworczak and Smolin (2026): trust regions; one-half threshold in binary states.

Competition among AI advisers

Collina et al. (2025, rev. 2026): strategic competition among different objectives can yield aligned outcomes.

Delegation under uncertain alignment

Fudenberg and Liang (2026): treatment-rate constraints and restrictions on input informativeness.

Here: aggregate common-evidence reports while allowing coordination.

Multi-sender cheap talk: Krishna and Morgan (2001); Battaglini (2002).

References and comparisons

Contributions

Extend single-adviser robust trust to committees with common evidence.

1. Trust threshold: invariance to committee size

No finite committee lowers the one-half threshold for positive robust value. Finite nondegenerate posteriors and positive local payoff curvature suffice.

2. Optimal aggregation: agreement and disagreement

Trust regions depend on the majority's lead; ties need separate treatment. A jointly feasible worst-case strategy establishes optimality for any finite committee.

3. Value of redundancy: hold evidence fixed

Quantify gains near the threshold and the rate at which larger committees recover the value of directly observing the common evidence.

Exact aggregation and redundancy: symmetric finite posteriors, quadratic loss.

Information

A binary state $\omega \in \{0, 1\}$ has prior $\mu_0 = \mathbb{P}(\omega = 1)$.

Common evidence

A common signal s induces the posterior

$$\mu = \mathbb{P}(\omega = 1 \mid s), \quad \mu \sim \tau, \quad \mathbb{E}_\tau[\mu] = \mu_0.$$

τ is the ex ante distribution of posterior beliefs. Its mean equals the prior (Bayes plausibility).

The support of τ is finite and nondegenerate: at least two possible beliefs.

Reports

Each adviser observes μ and submits a report m_i . The decision-maker observes $\mathbf{m} = (m_1, \dots, m_K)$, but not μ .

Alignment and reporting

$K \geq 1$ advisers have independent types $\theta_i \in \{A, M\}$, with $\mathbb{P}(\theta_i = A) = \alpha$, independent of (ω, μ) .

Who knows what

All advisers observe (μ, θ) , including every adviser's type. They do not observe ω separately. The decision-maker sees neither μ nor the types.

Admissible reporting strategies

Let $\beta(\cdot \mid \mu, \theta)$ denote the joint law of $\mathbf{m}$.

- Aligned (A): must report $m_i = \mu$.
- Misaligned (M): unrestricted messages, possibly truthful, with arbitrary joint randomization.

α measures alignment, not prediction accuracy.

Actions and payoffs

$\nu \in [0, 1]$ indexes the action optimal for belief ν . Let $U(\mu)$ be the optimal expected payoff if μ were known.

Payoff at the true posterior μ

In the rich action environment, U is differentiable and convex, and

$$W(\nu, \mu) = U(\nu) + U'(\nu)(\mu - \nu).$$

Thus $W(\mu, \mu) = U(\mu)$ and $W(\nu, \mu) \leq U(\mu)$.

Example: quadratic regret

For $U(x) = x^2$,

$$W(\nu, \mu) = 2\nu\mu - \nu^2, \quad U(\mu) - W(\nu, \mu) = (\nu - \mu)^2.$$

The general model does not yet impose this quadratic specification.

Decision problem

Before reports arrive, the decision-maker commits to a rule a . Given $\mathbf{m}$, the rule selects $\nu \sim a(\mathbf{m})$. Deterministic rules are included.

Worst-case expected gain

$$V_K = \sup_a \inf_{\beta} \mathbb{E}_{\tau, \theta, \beta, a}[W(\nu, \mu) - W(\mu_0, \mu)].$$

For each rule, take the worst admissible reporting strategy β . Then choose the rule with the highest guarantee.

The expectation averages over evidence, types, and randomization, with τ and α fixed.

Benchmark

Ignoring all reports and choosing $\nu = \mu_0$ gives zero gain, so $V_K \geq 0$.

Trust threshold

The posterior distribution τ has finite, nondegenerate support. The convex payoff function satisfies $U \in C^2$ and $U''(\mu_0) > 0$.

Theorem 1

For every finite $K \geq 1$,

$$V_K > 0 \iff \alpha > \frac{1}{2}.$$

Symmetry of τ is not required.

More advisers do not lower the threshold for positive robust value. They can still improve value above the threshold.

α is an ex ante type probability, not a realized majority condition.

Impossibility: $\alpha \leq 1/2$

Illustration: $\mathbb{P}(\mu = 0.2) = \mathbb{P}(\mu = 0.8) = 1/2$, so $\mu_0 = 0.5$.

Aligned advisers report μ . Misaligned advisers mix independently between μ and $1 - \mu$, so that, averaging over types,

$$\mathbb{P}(m_i = \mu \mid \mu) = \mathbb{P}(m_i = 1 - \mu \mid \mu) = \frac{1}{2}.$$

Reports ($K = 3$)	$\mathbb{P}(\mathbf{m} \mid \mu = 0.8)$	$\mathbb{P}(\mathbf{m} \mid \mu = 0.2)$
(0.8, 0.8, 0.2)	1/8	1/8
(0.8, 0.8, 0.8)	1/8	1/8

Every ordered profile has probability 2^{-K} under either posterior. Thus $\mathbb{E}_\beta[\mu \mid \mathbf{m}] = \mu_0$ and $V_K = 0$.

For general τ , a common random fake preserves the posterior mean.

[Impossibility proof details](#)

Positive value: $\alpha > 1/2$

Use only adviser 1 and clip the report, for sufficiently small $t > 0$:

$$a_t(m_1) = \text{clip}(m_1; \mu_0 - t, \mu_0 + t).$$

Write $C = U''(\mu_0)\mathbb{E}_\tau|\mu - \mu_0| > 0$.

Guaranteed improvement of this rule

$$\begin{aligned}\Delta(t) &= \underbrace{\alpha Ct}_{\text{aligned gain}} - \underbrace{(1 - \alpha)Ct}_{\text{worst damage}} + o(t) \\ &= (2\alpha - 1)Ct + o(t) > 0.\end{aligned}$$

Both effects are first order. Their probability weights determine the sign.

Hence $V_K \geq V_1 \geq \Delta(t) > 0$. Using one adviser need not be optimal.

Derivation of the local expansion

Quadratic loss and full-information value

From here: finite nondegenerate τ , symmetric about μ_0 , quadratic regret $(\nu - \mu)^2$, and $1/2 < \alpha < 1$.

Prior benchmark: ignore all reports

$$\text{Var}_\tau(\mu) := \mathbb{E}_\tau[(\mu - \mu_0)^2] = \sum_{u \in S} \tau(u)(u - \mu_0)^2, \quad S = \text{supp}(\tau).$$

This is the expected loss from choosing $\nu = \mu_0$.

Full information: observe the common posterior directly

Choose $\nu = \mu$, so regret is zero. The improvement over the prior is

$$V^{\text{FI}} := \text{Var}_\tau(\mu) - 0 = \text{Var}_\tau(\mu).$$

Observing μ does not reveal the realized state ω .

Benchmark derivation

Theorem 2: proof roadmap and value bounds

Define expected loss, including action randomization, by

$$L(a, \beta) := \mathbb{E}[(\nu - \mu)^2], \quad \nu \sim a(\mathbf{m}).$$

A rule's guaranteed value is $g(a) := V^{\text{FI}} - \sup_{\beta} L(a, \beta)$, and $V_K = \sup_a g(a)$.

Steps

What the proof must establish

- | | |
|--------------------------|--|
| 1–3. Candidate and radii | Calculate $g(a^*) = \bar{V}$, so $V_K \geq \bar{V}$. |
| 4. Fixed-attack bound | Even a best response to a given β yields only $I(\beta)$. |
| 5. Feasible attack | Construct one β^* with $I(\beta^*) = \bar{V}$. |
| 6. Matching bounds | Conclude $V_K = \bar{V}$ and global optimality. |

A fixed rule gives a lower bound. A fixed attack gives an upper bound.

Step 1: the candidate margin rule

Let $0 < t_1 < \dots < t_K$ be trust radii, specified in Step 3. Count identical report values.

Strict majority

If $c > K/2$ advisers report u , define the margin $d = 2c - K$ and choose

$$a^*(\mathbf{m}) = \text{clip}(u; \mu_0 - t_d, \mu_0 + t_d).$$

Tie and other profiles

Exactly two values, each reported $K/2$ times:

- If their interval contains μ_0 , choose μ_0 .
- Otherwise take the less extreme value, clipped at radius t_K .

All other profiles: choose μ_0 (including a 2–2–1 plurality).

For $K = 2$: clipped agreement, or clipped least-extreme disagreement.

Step 2: worst loss of the candidate rule

Fix a posterior μ and a type profile with j misaligned advisers. Write $x = |\mu - \mu_0|$ and $(y)_+ := \max\{y, 0\}$.

Proposition 1: conditional worst loss

$$L_j(\mu) = \begin{cases} (x - t_{K-2j})_+^2, & j < K/2, \\ x^2, & j = K/2, \\ (x + t_{2j-K})^2, & j > K/2. \end{cases}$$

- Truthful majority: keep its margin minimal, leaving clipping loss.
- Equal types: reflected reports induce the prior action.
- Misaligned majority: reach the trust boundary opposite the truth.

The bounds are attainable over all feasible reports.

[Proof of all three cases](#)

Step 2: from worst loss to guaranteed value

Type independence gives $w_{K,j} := \binom{K}{j} \alpha^{K-j} (1-\alpha)^j$. Write $\tilde{m} := |\mu - \mu_0|$ and $M := \mathbb{E}_\tau[\tilde{m}]$.

$$G(t) := \mathbb{E}_\tau[\tilde{m}^2 - (\tilde{m} - t)_+^2], \quad D(t) := \mathbb{E}_\tau[(\tilde{m} + t)^2 - \tilde{m}^2] = 2Mt + t^2.$$

G is the gain with a truthful majority; D is the damage with a false majority.

Average over evidence and types

$$\bar{L} := \sum_{j=0}^K w_{K,j} \mathbb{E}_\tau[L_j(\mu)], \quad \bar{V} := V^{\text{FI}} - \bar{L},$$

$$\bar{V} = \sum_{j < K/2} w_{K,j} G(t_{K-2j}) - \sum_{j > K/2} w_{K,j} D(t_{2j-K}).$$

Thus $g(a^*) = \bar{V} \leq V_K$. A better rule has not yet been excluded.

Step 3: the gain–damage balance fixes the radii

Set $c = (K + d)/2$, $\ell = (K - d)/2$ and fix a majority set B of size c . Write p_d^+ for B aligned and p_d^- for B misaligned:

$$p_d^+ = \alpha^c(1 - \alpha)^\ell, \quad p_d^- = \alpha^\ell(1 - \alpha)^c, \quad \frac{p_d^+}{p_d^-} = r^d, \quad r := \frac{\alpha}{1 - \alpha}.$$

Marginal truthful gain equals marginal adversarial damage

The paired contribution is $\Phi_d(t) := p_d^+ G(t) - p_d^- D(t)$. Its unique maximizer satisfies

$$r^d \mathbb{E}_\tau[(\tilde{m} - t_d)_+] = M + t_d.$$

A larger d gives a larger t_d .

These are type probabilities, not posterior odds after arbitrary reports. Optimizing radii alone does not establish global optimality.

Derivative, uniqueness and monotonicity

Step 4: a fixed attack gives a value upper bound

Lemma 1: best response to a known reporting mechanism

Fix a feasible β and define $\hat{\mu}_\beta(\mathbf{m}) := \mathbb{E}_\beta[\mu \mid \mathbf{m}]$. Under quadratic loss,

$$\mathbb{E}_\beta[(\nu - \mu)^2 \mid \mathbf{m}] = \text{Var}_\beta(\mu \mid \mathbf{m}) + (\nu - \hat{\mu}_\beta(\mathbf{m}))^2.$$

Thus the best response chooses $\nu = \hat{\mu}_\beta(\mathbf{m})$.

Total variance gives its value

$$I(\beta) := V^{\text{FI}} - \mathbb{E}_\beta[\text{Var}_\beta(\mu \mid \mathbf{m})] = \text{Var}_\beta(\hat{\mu}_\beta(\mathbf{m})).$$

Even knowing this attack cannot yield more than $I(\beta)$, so

$$g(a^*) \leq V_K \leq I(\beta) \leq V^{\text{FI}}.$$

Variance decomposition and bound direction

Step 5: an attack that makes clipping a best response

Fix $d < K$, a labeled majority B , and boundary $b = \mu_0 + t_d$. Consider reports $u > b$ on B and $v \leq \mu_0$ on B^c .

True posterior	Aligned set	Joint source mass
u	Majority B	q_u
v	Minority B^c	q_v

Target: posterior mean equals the clipping action

$$q_u(u - b) = q_v(b - v) \implies \mathbb{E}_{\beta^*}[\mu \mid \mathbf{m}] = b = a^*(\mathbf{m}).$$

The radius equation balances total probability-weighted deviations. An explicit transport converts this balance into feasible report probabilities.

Interior reports, unanimity and ties also satisfy $\mathbb{E}_{\beta^*}[\mu \mid \mathbf{m}] = a^*(\mathbf{m})$.

Full construction and all profile checks

Step 6: the saddle point proves Theorem 2

The same pair satisfies both best-response inequalities

$$\underbrace{L(a^*, \beta)}_{\text{any other attack}} \leq \underbrace{L(a^*, \beta^*)}_{\bar{L}} \leq \underbrace{L(a, \beta^*)}_{\text{any other rule}} .$$

Left: β^* attains every conditional worst loss in Proposition 1. Right: a^* is the posterior-mean action under β^* .

For every a , $\sup_{\beta} L(a, \beta) \geq L(a, \beta^*) \geq \bar{L}$, while a^* guarantees worst loss $\bar{L}$.

Theorem 2: exact value and global optimality

$$\inf_a \sup_{\beta} L(a, \beta) = \bar{L}, \quad V_K = V^{\text{FI}} - \bar{L} = \bar{V}.$$

The margin rule is optimal among all rules, including randomized and asymmetric rules.

Corollary 1: the exact value with two advisers

Set $K = 2$ in Theorem 2.

Types	Probability	Worst relative value
Both aligned	α^2	$G(t_2)$
One of each	$2\alpha(1 - \alpha)$	0
Both misaligned	$(1 - \alpha)^2$	$-D(t_2)$

An implication of Theorem 2

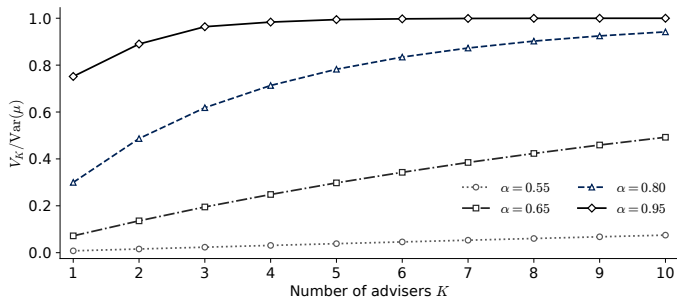
$$V_2 = \alpha^2 G(t_2) - (1 - \alpha)^2 D(t_2) = \alpha^2 \mathbb{E}[\min\{\tilde{m}, t_2\}^2] + (1 - \alpha)^2 t_2^2.$$

The radius equation gives the second equality, not a new bound.

Zero is the worst relative gain, not the value of every disagreement.

The two-adviser algebra

The value of redundancy



Vertical axis: $V_K / V^{\text{FI}} = V_K / \text{Var}_\tau(\mu)$, the fraction of available value.

Uniform posterior on $\{0, 1/8, \dots, 1\}$. Values from the exact formula.

Individual alignment affects the size of the gain

At $K = 10$: about 7.5% for $\alpha = .55$, compared with 94.2% for $\alpha = .80$.

Near-threshold value: fixed committee size

Fix K and τ . Let $\alpha = \frac{1}{2} + \varepsilon$, with $\varepsilon \downarrow 0$.

Corollary 2(i): relative value

$$\frac{V_K(\alpha)}{V_1(\alpha)} \longrightarrow K.$$

Both values vanish at order ε^2 despite the relative amplification.

Corollary 2(iii): local quality equivalent

Define $f(\alpha) > 1/2$ locally by $V_1(f(\alpha)) = V_2(\alpha)$. Then

$$f\left(\frac{1}{2} + \varepsilon\right) = \frac{1}{2} + \sqrt{2}\varepsilon + o(\varepsilon).$$

Only the distance above $1/2$ is scaled. The equivalence is local.

Local expansion and quality-equivalent proof

Large committees: fixed individual alignment

Fix $\alpha \in (1/2, 1)$ and τ . Let $K \rightarrow \infty$.

$$\frac{V_K}{V^{\text{FI}}} \longrightarrow 1, \quad R_K := V^{\text{FI}} - V_K, \quad \rho_K := \frac{R_K}{V^{\text{FI}}}.$$

R_K is remaining worst-case loss. ρ_K is the fraction of value still unrecovered.

Corollary 2(ii): the exact exponential rate

$$-\frac{1}{K} \log \rho_K \longrightarrow \lambda(\alpha), \quad \lambda(\alpha) := \text{KL}\left(\frac{1}{2} \parallel \alpha\right) = -\frac{1}{2} \log(4\alpha(1-\alpha)) > 0.$$

Natural logs. The same exponent governs failure of an aligned majority.

- No aligned majority: unavoidable loss limits how fast ρ_K can fall.
- Aligned majority: clipping loss must also shrink fast enough.

Matching loss bounds and the exponential rate

Conclusion and scope

Trust threshold

For every finite K , $V_K > 0$ if and only if $\alpha > 1/2$ under Theorem 1's assumptions. More advisers leave this threshold unchanged.

Optimal aggregation

A strict majority selects the report to follow. Its margin determines the trust radius. Extreme reports can still require clipping.

Value of redundancy

Above the threshold, more advisers amplify value and, as K grows, recover the full-information benchmark of observing the common posterior.

Common evidence, independent types, unrestricted coordinated reports. Exact aggregation and redundancy require symmetric posteriors and quadratic loss.

Truthful-report probability

Suppose $\alpha \leq 1/2$. Fix a truth $u \neq \mu_0$ and a common fake Z . A misaligned adviser reports u with probability γ , otherwise Z .

$$\mathbb{P}(m_i = u \mid u, Z) = \alpha + (1 - \alpha)\gamma.$$

Setting this probability to $1/2$ gives

$$\gamma = \frac{1/2 - \alpha}{1 - \alpha} = \frac{1 - 2\alpha}{2(1 - \alpha)} \in [0, 1/2].$$

For example, $\alpha = 0.4$ gives $\gamma = 1/6$ and $0.4 + 0.6/6 = 0.5$.

Independent types and private coins give probability 2^{-K} to each ordered vector in $\{u, Z\}^K$, conditional on (u, Z) . Misalignment permits truth-telling, so simply reversing reports cannot help.

Asymmetric posteriors: the fake kernel

Bayes plausibility balances total deviations on the two sides:

$$S_\tau = \sum_{u > \mu_0} \tau(u)(u - \mu_0) = \sum_{v < \mu_0} \tau(v)(\mu_0 - v) > 0.$$

For $u > \mu_0 > v$, choose the common fake using

$$Q(v | u) = \frac{\tau(v)(\mu_0 - v)}{S_\tau}, \quad Q(u | v) = \frac{\tau(u)(u - \mu_0)}{S_\tau}.$$

Each row sums to one. At $\mu = \mu_0$, all reports equal μ_0 .

Pairwise cancellation

$$\tau(u)Q(v | u)(u - \mu_0) = \tau(v)Q(u | v)(\mu_0 - v).$$

Opposing explanations need equal weighted deviations, not equal probabilities.

Asymmetric posteriors: mixed reports

Fix an ordered report profile containing exactly $u > \mu_0 > v$. Only two truth-fake pairs can generate it:

$$(\mu, Z) = (u, v) \quad \text{or} \quad (v, u).$$

Their joint masses, after averaging over types and private coins, are

$$q_u = 2^{-K} \tau(u) Q(v \mid u), \quad q_v = 2^{-K} \tau(v) Q(u \mid v).$$

Therefore

$$\mathbb{E}_\beta[\mu \mid \mathbf{m}] - \mu_0 = \frac{q_u(u - \mu_0) - q_v(\mu_0 - v)}{q_u + q_v} = 0.$$

The numerator vanishes by pairwise cancellation. The denominator is positive for an on-path profile.

Asymmetric posteriors: unanimity

Consider $\mathbf{m} = (u, \dots, u)$ with $u > \mu_0$.

- Truth u : all coordinates select truth, whatever the fake $v < \mu_0$.
- Truth $v < \mu_0$: the common fake is u and all coordinates select it.

Removing the common factor 2^{-K} , the posterior-deviation numerator is

$$\begin{aligned} & \tau(u)(u - \mu_0) + \sum_{v < \mu_0} \tau(v)Q(u | v)(v - \mu_0) \\ &= \sum_{v < \mu_0} \left[\tau(u)Q(v | u)(u - \mu_0) + \tau(v)Q(u | v)(v - \mu_0) \right] = 0. \end{aligned}$$

The first equality uses $\sum_{v < \mu_0} Q(v | u) = 1$.

The case $u < \mu_0$ is analogous. A unanimous μ_0 report comes only from truth μ_0 .

These and mixed reports exhaust all on-path profiles.

Why a constant posterior mean implies zero value

Fix the constructed β and an on-path $\mathbf{m}$. With $\hat{\mu} = \mathbb{E}_\beta[\mu \mid \mathbf{m}]$, linearity in μ gives

$$\begin{aligned}\mathbb{E}_\beta[W(\nu, \mu) \mid \mathbf{m}] &= U(\nu) + U'(\nu)(\hat{\mu} - \nu) \\ &= W(\nu, \hat{\mu}) \leq U(\hat{\mu}).\end{aligned}$$

If $\hat{\mu} = \mu_0$ at every profile, choosing $\nu = \mu_0$ attains this bound everywhere. Randomization cannot improve it.

Even knowing this attack cannot improve on the prior action. Thus $V_K \leq 0$; ignoring reports gives $V_K \geq 0$, hence $V_K = 0$.

This argument uses general convex U . The reports may still reveal features of μ beyond its mean.

Positive value: local payoff expansion

Let $S = \text{supp}(\tau)$ and $\delta = \min_{u \in S, u \neq \mu_0} |u - \mu_0| > 0$. Choose $0 < t < \min\{\delta, \mu_0, 1 - \mu_0\}$, small enough that $U'' > 0$ throughout $[\mu_0 - t, \mu_0 + t]$.

Differentiate the payoff

$$\begin{aligned}\partial_\nu W(\nu, u) &= U'(\nu) + U''(\nu)(u - \nu) - U'(\nu) \\ &= U''(\nu)(u - \nu).\end{aligned}$$

Integrating from μ_0 to $\mu_0 + h$, for $|h| \leq t$,

$$W(\mu_0 + h, u) - W(\mu_0, u) = U''(\mu_0)(u - \mu_0)h + o(t).$$

Continuity of U'' gives a uniform remainder over this interval and finite S .

Positive value: aligned gain

Fix $u \neq \mu_0$ and let $s_u = \text{sign}(u - \mu_0)$. Since $t < \delta$, a truthful report induces $a_t(u) = \mu_0 + s_u t$.

$$\begin{aligned} W(\mu_0 + s_u t, u) - W(\mu_0, u) &= \int_0^t U''(\mu_0 + s_u x)(|u - \mu_0| - x) dx \\ &= U''(\mu_0)|u - \mu_0|t + o(t). \end{aligned}$$

The action moves toward u , so its first-order gain is positive.

Averaging over the evidence

At $u = \mu_0$, a truthful report leaves the action unchanged. Therefore

$$A(t) := \mathbb{E}_\tau[W(a_t(\mu), \mu) - W(\mu_0, \mu)] = Ct + o(t),$$

where $C = U''(\mu_0)\mathbb{E}_\tau|\mu - \mu_0| > 0$.

Positive value: worst damage and net gain

For $u \neq \mu_0$, $\partial_\nu W = U''(\nu)(u - \nu)$ has constant sign on the trust interval. The worst action is $\mu_0 - s_u t$.

$$\min_{|h| \leq t} \{W(\mu_0 + h, u) - W(\mu_0, u)\} = -U''(\mu_0)|u - \mu_0|t + o(t).$$

Misaligned reports can reach that endpoint. At $u = \mu_0$, the worst change is $O(t^2)$, so averaging gives $B(t) = -Ct + o(t)$.

$$\begin{aligned}\Delta(t) &= \alpha A(t) + (1 - \alpha)B(t) \\ &= (2\alpha - 1)Ct + o(t).\end{aligned}$$

If $\alpha > 1/2$, set $c = (2\alpha - 1)C > 0$. For small enough t , $|o(t)| \leq ct/2$, so $\Delta(t) \geq ct/2 > 0$.

Benchmarks: loss, value and full information

Under quadratic regret, subtracting prior payoff gives

$$W(\nu, \mu) - W(\mu_0, \mu) = (\mu_0 - \mu)^2 - (\nu - \mu)^2.$$

Taking expectations under a fixed rule and attack,

$$\mathbb{E}[W(\nu, \mu) - W(\mu_0, \mu)] = V^{\text{FI}} - L(a, \beta).$$

Why the optimization direction changes

$$g(a) = \inf_{\beta} \{V^{\text{FI}} - L(a, \beta)\} = V^{\text{FI}} - \sup_{\beta} L(a, \beta),$$

$$V_K = \sup_a g(a) = V^{\text{FI}} - \inf_a \sup_{\beta} L(a, \beta).$$

Ignoring reports guarantees zero value; nonnegative loss gives $0 \leq V_K \leq V^{\text{FI}}$.

For equally likely $\mu = 0.2, 0.8$, $\mu_0 = 0.5$: $V^{\text{FI}} = \frac{1}{2}(0.3)^2 + \frac{1}{2}(0.3)^2 = 0.09$.

Lemma 1: the variance decomposition and upper bound

Let $\hat{\mu} = \mathbb{E}_{\beta}[\mu \mid \mathbf{m}]$. Expand $\nu - \mu = (\nu - \hat{\mu}) + (\hat{\mu} - \mu)$. The conditional cross term vanishes because $\mathbb{E}_{\beta}[\hat{\mu} - \mu \mid \mathbf{m}] = 0$.

$$\mathbb{E}_{\beta}[(\nu - \mu)^2 \mid \mathbf{m}] = \text{Var}_{\beta}(\mu \mid \mathbf{m}) + (\nu - \hat{\mu})^2.$$

Randomization averages a nonnegative second term, so $\nu = \hat{\mu}$ minimizes loss.

Recovered value and the bound direction

$$V^{\text{FI}} = \mathbb{E}_{\beta}[\text{Var}_{\beta}(\mu \mid \mathbf{m})] + \text{Var}_{\beta}(\hat{\mu}), \quad I(\beta) = \text{Var}_{\beta}(\hat{\mu}).$$

For a fixed feasible β_0 and every a ,

$$g(a) \leq V^{\text{FI}} - L(a, \beta_0) \leq \sup_{\tilde{a}} \{V^{\text{FI}} - L(\tilde{a}, \beta_0)\} = I(\beta_0).$$

Taking $\sup_a$ gives $V_K \leq I(\beta_0)$, then $V_K \leq \inf_{\beta} I(\beta)$.

Proposition 1: truthful majority and equal types

Fix μ , a labeled type profile, and $x = |\mu - \mu_0|$.

If $j < K/2$: truthful majority

The $K - j$ aligned advisers report μ . Its report count is at least $K - j$, so its margin is at least $K - 2j$. Increasing radii imply an action between the prior and truth, with

$$(a^*(\mathbf{m}) - \mu)^2 \leq (x - t_{K-2j})_+^2.$$

Equality: every misaligned adviser reports the same value different from μ .

If $j = K/2$: equal numbers of both types

Joining the truth creates a truthful majority. A common different report gives a two-bloc tie. Split false reports leave no majority. Every case selects an action between μ_0 and μ , so loss is at most x^2 . Reflection $2\mu_0 - \mu$ attains x^2 .

Proposition 1: misaligned majority

Suppose $j > K/2$. A fake strict majority has at most j members, so its margin is at most $2j - K$ and its radius at most t_{2j-K} .

- A truthful majority acts between the prior and truth.
- Other no-majority profiles return to the prior.
- A two-bloc tie cannot go beyond the truth on its side or cross the prior: the truthful report is one of the two bloc values.

Thus every report pattern gives loss at most $(x + t_{2j-K})^2$. A common false report beyond the opposite boundary attains this bound. If all advisers are misaligned, the rule always stays within radius t_K , including on ties. The same bound holds.

Proposition 1: averaging the attainable losses

The adversary observes μ and the entire type profile. It can therefore choose an attaining report separately for every conditional event.

Conditional attainability becomes the guarantee

$$\sup_{\beta} L(a^*, \beta) = \sum_{j=0}^K w_{K,j} \mathbb{E}_{\tau}[L_j(\mu)] = \bar{L}, \quad g(a^*) = V^{\text{FI}} - \bar{L}.$$

Subtract each conditional loss from $\tilde{m}^2$:

$$\begin{array}{l|l} j < K/2 & \mathbb{E}[\tilde{m}^2 - (\tilde{m} - t_{K-2j})_+^2] = G(t_{K-2j}) \\ j = K/2 & \mathbb{E}[\tilde{m}^2 - \tilde{m}^2] = 0 \\ j > K/2 & \mathbb{E}[\tilde{m}^2 - (\tilde{m} + t_{2j-K})^2] = -D(t_{2j-K}). \end{array}$$

Weighting these terms by $w_{K,j}$ gives $\bar{V}$. Since $V_K = \sup_a g(a)$, $\bar{V}$ is a lower bound on the optimal value.

Trust radii: differentiating the paired value

Fix B with $c = (K + d)/2$ members, and $\ell = K - c$. Its two complementary type events have probabilities p_d^+ and p_d^- .

$$\frac{p_d^+}{p_d^-} = \alpha^{c-\ell}(1-\alpha)^{\ell-c} = r^d.$$

With $\Phi_d(t) = p_d^+ G(t) - p_d^- D(t)$, differentiation gives

$$G'(t) = 2\mathbb{E}[(\tilde{m} - t)_+], \quad D'(t) = 2(M + t),$$

$$\Phi'_d(t) = 2p_d^+ \mathbb{E}[(\tilde{m} - t)_+] - 2p_d^-(M + t).$$

The first-order condition

$$\Phi'_d(t_d) = 0 \quad \Longleftrightarrow \quad r^d \mathbb{E}[(\tilde{m} - t_d)_+] = M + t_d.$$

$\Phi_d(0) = 0$ and $\Phi'_d(0) = 2(p_d^+ - p_d^-)M > 0$. Hence the maximizing radius gives a positive paired contribution.

Trust radii: existence, uniqueness and monotonicity

Define $\bar{m} := \max_{\mu \in S} |\mu - \mu_0|$ and

$$F_d(t) := r^d \mathbb{E}[(\bar{m} - t)_+] - (t + M), \quad 0 \leq t \leq \bar{m}.$$

F_d is continuous and strictly decreasing, because its first term is weakly decreasing and $-(t + M)$ is strictly decreasing.

$$F_d(0) = (r^d - 1)M > 0, \quad F_d(\bar{m}) = -(\bar{m} + M) < 0.$$

Thus it has exactly one root $t_d \in (0, \bar{m})$.

Increasing the margin increases the root

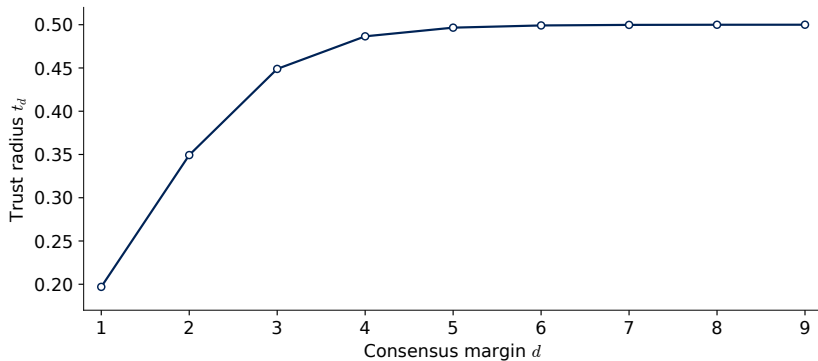
For $d' > d$,

$$F_{d'}(t_d) = (r^{d'} - r^d) \mathbb{E}[(\bar{m} - t_d)_+] > 0.$$

Strict decrease of $F_{d'}$ implies $t_{d'} > t_d$.

Symmetric support in $[0, 1]$ gives $\bar{m} \leq \min\{\mu_0, 1 - \mu_0\}$, so all trust intervals are feasible action intervals.

Trust radii and consensus margins



$$t_{d+1} > t_d, \quad t_d \longrightarrow \bar{m}.$$

Uniform posterior on $\{0, 1/8, \dots, 1\}$, $\alpha = 0.80$.

Saddle construction: objects and proof obligations

Fix a feasible $d > 0$, $c = (K + d)/2$, $\ell = (K - d)/2$, and a labeled set B of size c .

$$p_d^+ = \alpha^c(1 - \alpha)^\ell, \quad p_d^- = \alpha^\ell(1 - \alpha)^c, \quad b = \mu_0 + t_d.$$

Two complementary reporting events

Truth $u > b$, with B aligned: the minority draws a common $v \leq \mu_0$ using $\sigma(v \mid u)$.

Truth $v \leq \mu_0$, with B misaligned: the majority draws a common $u > b$ using $\kappa(u \mid v)$.

1. Balance the total deviations above and below b .
2. Convert the balance into probability rows summing to one.
3. Check posterior means for every on-path report category.
4. Verify that the same attack attains all conditional worst losses.

Saddle construction: the radius balances total drag

Drag means probability mass times distance from the target mean b . Define demand for $u > b$ and supply for $v < \mu_0$:

$$\mathcal{D}(u) := p_d^+ \tau(u)(u - b), \quad \mathcal{S}(v) := p_d^- \tau(v)(b - v).$$

If $\tau(\mu_0) > 0$, assign half to this side: $\mathcal{S}(\mu_0) := \frac{1}{2} p_d^- \tau(\mu_0) t_d$.

Symmetry and the radius equation

$$\sum_{u>b} \mathcal{D}(u) = \frac{p_d^+}{2} \mathbb{E}[(\tilde{m} - t_d)_+], \quad \sum_{v \leq \mu_0} \mathcal{S}(v) = \frac{p_d^-}{2} (t_d + M).$$

These totals are equal because $p_d^+ / p_d^- = r^d$. Denote their positive common value by T .

For supply, use $\Pr(\mu < \mu_0) = (1 - \tau(\mu_0))/2$ and $\sum_{v < \mu_0} \tau(v)(\mu_0 - v) = M/2$. Adding half the prior atom supplies the missing $\tau(\mu_0)t_d/2$.

Saddle construction: transport gives valid probabilities

Set

$$z(v, u) := \frac{\mathcal{S}(v)\mathcal{D}(u)}{T}, \quad \sum_u z(v, u) = \mathcal{S}(v), \quad \sum_v z(v, u) = \mathcal{D}(u).$$

Define the conditional draws

$$\sigma(v | u) := \frac{z(v, u)}{\mathcal{D}(u)}, \quad \kappa(u | v) := \frac{z(v, u)}{p_d^- \tau(v)(b - v)}.$$

Every state–type row has total probability one

$$\sum_v \sigma(v | u) = 1, \quad \sum_u \kappa(u | v) = \begin{cases} 1, & v < \mu_0, \\ 1/2, & v = \mu_0. \end{cases}$$

Reflect the construction around μ_0 for the lower boundary. The two half-rows at the prior sum to one.

Nonnegative entries preserve τ , type probabilities and all aligned reports.

Saddle construction: one strategy for all type profiles

Apply the following rules separately to every labeled aligned set.

Aligned majority B , with margin d

Exterior truth $u > \mu_0 + t_d$: minority draws a common fake via σ . Use the reflected kernel for $u < \mu_0 - t_d$.

Interior truth $|u - \mu_0| \leq t_d$: minority reports a unique tag $\eta_{B,u} \in [0, 1] \setminus S$. All-aligned profiles simply report u .

Misaligned majority B , with margin d

At low truth, draw a common exterior high fake via κ . At high truth, use the reflected low-side kernel. At the prior, combine the two half-rows.

If K is even and types split equally, every misaligned adviser reports $\rho(u) := 2\mu_0 - u$.

Best response: exterior profiles with a nonempty minority

For $d < K$, observe $u > b$ on B and $v \leq \mu_0$ on B^c . Under this single β^* , exactly two events produce this profile:

$$q_u = p_d^+ \tau(u) \sigma(v | u) = \frac{z(v, u)}{u - b},$$
$$q_v = p_d^- \tau(v) \kappa(u | v) = \frac{z(v, u)}{b - v}.$$

Therefore $q_u(u - b) = z(v, u) = q_v(b - v)$, and

$$\mathbb{E}_{\beta^*}[\mu | \mathbf{m}] = \frac{q_u u + q_v v}{q_u + q_v} = b = a^*(\mathbf{m}).$$

Illustration: reports (0.8, 0.8, 0.2)

Equal prior probabilities on $\mu = 0.2, 0.8$, $K = 3$, $\alpha = 0.75$: $r = 3$, $d = 1$, and $3(0.3 - t_1) = 0.3 + t_1$, so $t_1 = 0.15$. The source-mass ratio is 3 : 1, giving $(3 \cdot 0.8 + 0.2)/4 = 0.65$.

Best response: interior profiles, ties and source separation

Interior truthful majority

A unique minority tag identifies (B, u) . Thus the posterior mean is u . Since $|u - \mu_0| \leq t_d$, the margin action is also u . Its conditional worst loss is zero, so revealing u does not help against the bound.

Equal-type ties

Truth u with B aligned and truth $\rho(u)$ with B^c aligned have equal type probabilities and $\tau(u) = \tau(\rho(u))$. Their mean is $(u + \rho(u))/2 = \mu_0$, the rule's action.

Nonunanimous exterior profiles identify d by multiplicities and B by coordinates. Tags are outside S , so tagged profiles cannot collide with exterior profiles or ties.

At $u = \mu_0$, collapsed prior profiles still have posterior mean μ_0 .

Best response: unanimity requires summing over sources

Set $d = K$, $p_K^+ = \alpha^K$, $p_K^- = (1 - \alpha)^K$ and $b = \mu_0 + t_K$. At the unanimous exterior report $(u, \dots, u)$, $u > b$:

$$q_u = p_K^+ \tau(u), \quad q_v = p_K^- \tau(v) \kappa(u | v) = \frac{z(v, u)}{b - v}.$$

The opposite source v is unobserved, so all such sources must be added.

The aggregate posterior-deviation numerator is zero

$$q_u(u - b) - \sum_{v \leq \mu_0} q_v(b - v) = \mathcal{D}(u) - \sum_{v \leq \mu_0} z(v, u) = 0.$$

Hence $\mathbb{E}_{\beta^*}[\mu | (u, \dots, u)] = b$. The lower exterior case is symmetric.

A unanimous interior $u \neq \mu_0$ comes only from its all-aligned truth. At $u = \mu_0$, any collapsed prior tie has the same mean μ_0 .

The saddle certificate: both inequalities for the same pair

The constructed β^* attains Proposition 1 condition by condition:

- True exterior majority stays at its smallest margin; interior loss is zero.
- False majority reaches the opposite boundary, including at the prior.
- Equal types use reflection and attain prior loss.

Thus $L(a^*, \beta) \leq L(a^*, \beta^*) = \bar{L}$ for every β . The profile checks give $a^* = \mathbb{E}_{\beta^*}[\mu \mid \mathbf{m}]$ on path, so $L(a^*, \beta^*) \leq L(a, \beta^*)$ for every a .

The minimax loss is pinned down without interchanging operators

$$\bar{L} \leq \inf_a \sup_{\beta} L(a, \beta) \leq \sup_{\beta} L(a^*, \beta) = \bar{L}.$$

Consequently $I(\beta^*) = V^{\text{FI}} - \bar{L} = \bar{V} = V_K$. The inequalities cover randomized and asymmetric rules.

Corollary 1: the two-adviser algebra

Put $A = \alpha^2$, $B = (1 - \alpha)^2$, $t = t_2$. Theorem 2 gives $V_2 = AG(t) - BD(t)$; the radius equation gives

$$A\mathbb{E}[(\tilde{m} - t)_+] = B(t + M).$$

The pointwise identity

$$\tilde{m}^2 - (\tilde{m} - t)_+^2 = \min\{\tilde{m}, t\}^2 + 2t(\tilde{m} - t)_+$$

holds for both $\tilde{m} \leq t$ and $\tilde{m} > t$.

Substitution and cancellation

$$\begin{aligned} V_2 &= A\mathbb{E}[\min\{\tilde{m}, t\}^2] + 2At\mathbb{E}[(\tilde{m} - t)_+] - B(2Mt + t^2) \\ &= A\mathbb{E}[\min\{\tilde{m}, t\}^2] + 2Bt(t + M) - B(2Mt + t^2) \\ &= A\mathbb{E}[\min\{\tilde{m}, t\}^2] + Bt^2. \end{aligned}$$

This is an equality only after Theorem 2 establishes optimality.

Near-threshold proof: the value of a complementary pair

Fix K and τ . All $t_d \rightarrow 0$ as $\alpha \downarrow 1/2$. Put $p_0 := \Pr(\tilde{m} > 0) = 1 - \tau(\mu_0)$, excluding the prior atom.

Small radii remove the positive-part kink

Let $m_{\min}$ be the smallest positive deviation. For $0 \leq t < m_{\min}$,

$$\mathbb{E}[(\tilde{m} - t)_+] = M - p_0 t, \quad G(t) = 2Mt - p_0 t^2, \quad D(t) = 2Mt + t^2.$$

For $j < K/2$, put $d = K - 2j$ and define the complementary type-event probabilities

$$A_j = \binom{K}{j} \alpha^{K-j} (1 - \alpha)^j, \quad B_j = \binom{K}{j} \alpha^j (1 - \alpha)^{K-j}.$$

Their contribution is $2M(A_j - B_j)t - (A_j p_0 + B_j)t^2$. Maximizing gives

$$t_d = \frac{M(A_j - B_j)}{A_j p_0 + B_j}, \quad A_j G(t_d) - B_j D(t_d) = \frac{M^2(A_j - B_j)^2}{A_j p_0 + B_j}.$$

Near-threshold proof: the squared-margin identity

Let $\alpha = 1/2 + \varepsilon$ and $q_j = \binom{K}{j} 2^{-K}$. For fixed K, j ,

$$A_j - B_j = 4q_j d \varepsilon + O(\varepsilon^2), \quad A_j p_0 + B_j = q_j(1 + p_0) + O(\varepsilon).$$

The paired value is therefore $\frac{16M^2}{1+p_0} q_j d^2 \varepsilon^2 + o(\varepsilon^2)$.

The second moment of the margin

Independent auxiliary signs $X_i = \pm 1$ have equal probabilities.

$$\mathbb{E}[(\sum_i X_i)^2] = \sum_i \mathbb{E}[X_i^2] + 2 \sum_{i < l} \mathbb{E}[X_i X_l] = K.$$

Symmetry gives $\sum_{j < K/2} q_j (K - 2j)^2 = K/2$. The zero margin contributes zero.

Summing the finitely many paired values yields

$$V_K(1/2 + \varepsilon) = \frac{8KM^2}{1+p_0} \varepsilon^2 + o(\varepsilon^2), \quad \frac{V_K}{V_1} \rightarrow K.$$

Near-threshold proof: the local quality equivalent

Set $C_\tau = 8M^2/(1 + p_0) > 0$. Define the positive local solution by $V_1(f(\alpha)) = V_2(\alpha)$ and write $f(1/2 + \varepsilon) = 1/2 + x_\varepsilon$.

The two values share the same quadratic coefficient

$$C_\tau x_\varepsilon^2 + o(x_\varepsilon^2) = 2C_\tau \varepsilon^2 + o(\varepsilon^2).$$

The single-adviser local value is continuous and strictly increasing above $1/2$, selecting a unique small $x_\varepsilon > 0$. Hence

$$(C_\tau + o(1)) \frac{x_\varepsilon^2}{\varepsilon^2} = 2C_\tau + o(1), \quad x_\varepsilon = \sqrt{2} \varepsilon + o(\varepsilon).$$

For example, the local approximation at $\alpha = .51$ gives $f(.51) \approx .5 + \sqrt{2}(.01) = .51414$.

The approximation concerns the excess alignment probability above $1/2$. It supplies neither an exact equality at $.51$ nor a global quality–quantity ranking.

Large-committee proof: a lower loss bound

Fix $\alpha \in (1/2, 1)$ and τ . Let $q = 1 - \alpha$ and $J \sim \text{Binomial}(K, q)$ count misaligned advisers.

Failure of an aligned majority leaves unavoidable loss

Proposition 1 gives prior loss at a tie and at least prior loss when $J > K/2$.

$$R_K \geq V^{\text{FI}} \Pr(J \geq K/2), \quad \rho_K \geq \Pr(J \geq K/2).$$

Define $\text{KL}(x \| q) = x \log(x/q) + (1-x) \log((1-x)/(1-q))$ and $\lambda = \text{KL}(1/2 \| q) = \text{KL}(1/2 \| \alpha)$. With $x_K = \lceil K/2 \rceil / K$,

$$\frac{e^{-K \text{KL}(x_K \| q)}}{K+1} \leq \Pr(J = \lceil K/2 \rceil) \leq \Pr(J \geq K/2) \leq e^{-K\lambda}.$$

The bounds and $x_K \rightarrow 1/2$ give the binomial-tail rate λ . Therefore

$$\limsup_{K \rightarrow \infty} -\frac{1}{K} \log \rho_K \leq \lambda.$$

Remaining loss cannot decay faster than the majority-failure probability.

Large-committee proof: an upper loss bound

Set $\bar{m} = \max \tilde{m}$, $p_{\max} = \Pr(\tilde{m} = \bar{m}) > 0$ and $r = \alpha/(1 - \alpha) > 1$. Finite support ensures positive endpoint mass.

The radius equation controls truthful-majority clipping

$$p_{\max}(\bar{m} - t_d) \leq \mathbb{E}[(\tilde{m} - t_d)_+] = r^{-d}(t_d + M) \leq (\bar{m} + M)r^{-d}.$$

Thus, with $C = ((\bar{m} + M)/p_{\max})^2$ independent of K, d ,

$$\mathbb{E}[(\tilde{m} - t_d)_+^2] \leq (\bar{m} - t_d)^2 \leq Cr^{-2d}.$$

Bad-or-tied conditional loss is at most 4, since $\tilde{m}, t_d \leq 1$. Average over all possible margins, retaining $w_{K,j} = \binom{K}{j} \alpha^{K-j} (1 - \alpha)^j$:

$$R_K \leq 4 \Pr(J \geq K/2) + C \sum_{j < K/2} w_{K,j} r^{-2(K-2j)}.$$

The sum retains every margin, including small aligned majorities.

Large-committee proof: the two rates match

For $x = j/K \leq 1/2$, the binomial bound $w_{K,j} \leq e^{-K \text{KL}(x||1-\alpha)}$ gives

$$w_{K,j} r^{-2(K-2j)} \leq e^{-KH(x)}, \quad H(x) = \text{KL}(x||1-\alpha) + 2(1-2x) \log r.$$

$H''(x) = 1/[x(1-x)] > 0$ and $H'(1/2) = -3 \log r < 0$. Hence H decreases up to $1/2$ and $H(x) \geq H(1/2) = \lambda$.

Polynomially many margins cannot slow the exponent

There are at most $K+1$ terms, so the upper loss bound implies

$$\rho_K \leq \frac{4 + C(K+1)}{V^{\text{FI}}} e^{-K\lambda}, \quad \liminf_{K \rightarrow \infty} -\frac{1}{K} \log \rho_K \geq \lambda.$$

Combine this with the lower loss bound's $\limsup \leq \lambda$:

$$-\frac{1}{K} \log \rho_K \longrightarrow \lambda = -\frac{1}{2} \log(4\alpha(1-\alpha)) > 0, \quad \frac{V_K}{V^{\text{FI}}} \longrightarrow 1.$$

Scope of the results

Maintained assumptions

Common posterior evidence, independent adviser types, unrestricted misaligned reports, and rich actions. Misaligned advisers observe μ and the full type profile, but not ω separately.

Extensions outside the established results

- Independent signals or verifiable reports.
- Correlated types or additional state information.
- Restricted action sets.
- Asymmetric posteriors in the exact-value problem.

Numerical verification

The deterministic audit covers $K = 1, \dots, 6$ and $\alpha \in \{.55, .65, .80, .95\}$.

Quantity	Maximum error
Conditional probability row	$< 9 \times 10^{-13}$
Posterior action / pointwise loss	$< 2.3 \times 10^{-16}$
Aggregate formula	$< 1.1 \times 10^{-14}$
Independent LP comparison ($K \leq 3$)	$< 1.4 \times 10^{-16}$

These calculations check the analytical construction. The arbitrary- K optimality claim follows from the proof.

Related AI literature: mechanisms and scope

Dworczak and Smolin (2026)

Robust Trust. arXiv:2602.09490v2, March 19. Foundation: single-adviser trust regions, the binary half-threshold and saddle-point logic. Extension: jointly generated committee reports.

Collina, Goel, Roth, Ryu and Shi (2025; revised 2026)

Emergent Alignment via Competition. arXiv:2509.15090v2, February 2. Equilibrium gains from competing objectives, with a convex-hull condition on user utility. Here misaligned reports may coordinate: the guarantee does not rely on competition.

Fudenberg and Liang (2026)

Friend or Foe: Delegating to an AI whose Alignment is Unknown. June 22. Constrain treatment rates and input informativeness. Here: aggregate reports about fixed evidence.

Multiple advisers: the distinction from cheap talk

Classic multi-sender communication

Krishna, V., and J. Morgan (2001). *A Model of Expertise*. QJE 116(2), 747–775.

Battaglini, M. (2002). *Multiple Referrals and Multidimensional Cheap Talk*. *Econometrica* 70(4), 1379–1401.

Different restrictions on strategic behavior

These models study equilibrium information transmission under specified preferences and communication protocols. Our rule must withstand every admissible joint reporting law, subject to aligned advisers reporting truthfully.

The committee contribution

Finite-committee threshold invariance, majority-margin trust with tie handling, a joint saddle and redundancy values under common evidence.